Complete AI Automation & Agent with n8n

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Module 1

Automation

Module 1

What are Automations ?

Automation is the process of using technology to **perform tasks** without human intervention, making repetitive or time-consuming activities faster and more efficient.

Automations:

- Follow a fixed, rule-based process.
- Great for repetitive, predictable tasks.
- Example: An email sent to every new customer as soon as they sign up.

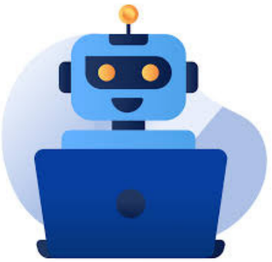
Automation are Deterministic and Predefined

Example : Automatically sending a welcome email to a new subscriber after they sign up for a newsletter.

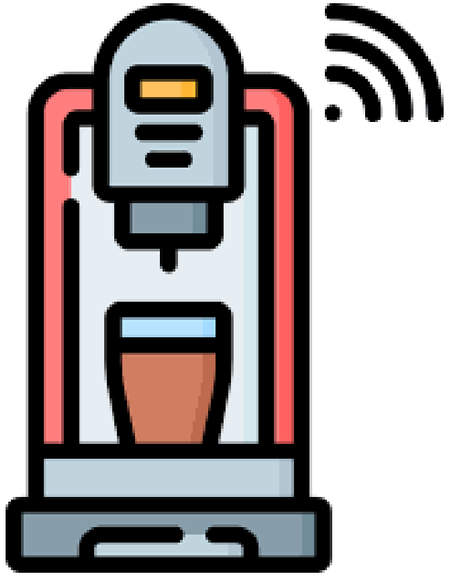
A machine that makes coffee at 7:00 AM every day.

Module 1

What are Agents ?



An AI agent is a node that uses artificial intelligence to perform smart tasks, like analyzing text or making decisions.



An AI agent is like a super-smart assistant. It’s the one who reads the room (or data) and figures out what to do next.

Flow

Module 2

n8n as an Automation Tool

What is n8n ?

N8N is an open-source, no-code tool that lets you connect apps and services to automate workflows and integrate AI-driven tasks.

We can start from free and it can also be hosted locally or on cloud

Module 2

n8n Foundations

N8N is an open-source, no-code tool that lets you connect apps and services to automate workflows and integrate AI-driven tasks.

We can start from free and it can also be hosted locally or on cloud

It has more than 500+ integrations with common applications and AI.

Module 2

n8n Foundations

Since this is a no-code tool, we can easily use and master it to build our automation and agents.

It’s free to start, beginner-friendly, and packed with integrations. No coding required—just drag, drop, and connect.

There’s a huge N8N community with templates and tutorials to guide you."

Module 2

n8n Foundations

- n8n Sign-up Process
- n8n UI Walkthrough
- n8n Workflow
- n8n first Automation

Ways to run n8n

1. Directly on n8n. (Free trial 14 days) + Monthly Pricing
2. Local.
3. Self hosted on a server.

Module 3

Nodes in n8n

What are Nodes?

A **Node** is a single building block in an N8N workflow, each performing a specific job like triggering an event or processing data.

Automation Practical

Let us see some practical use cases and build them in n8n from scratch.

Module 4

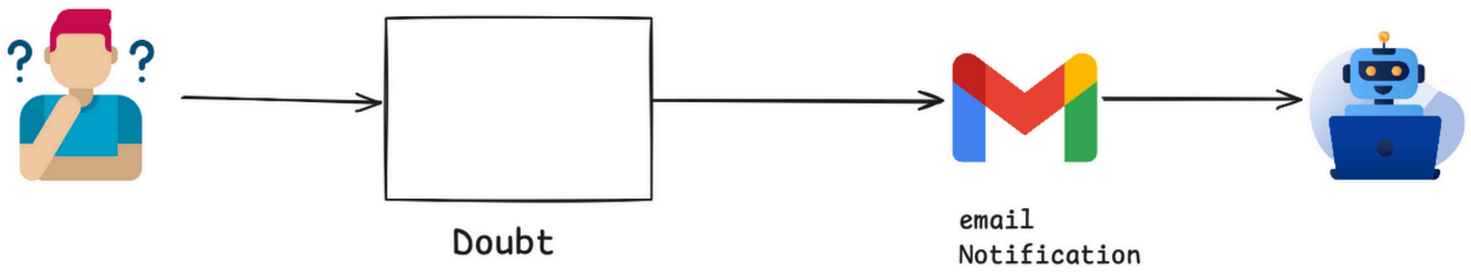
AI & Agents in n8n

Flow

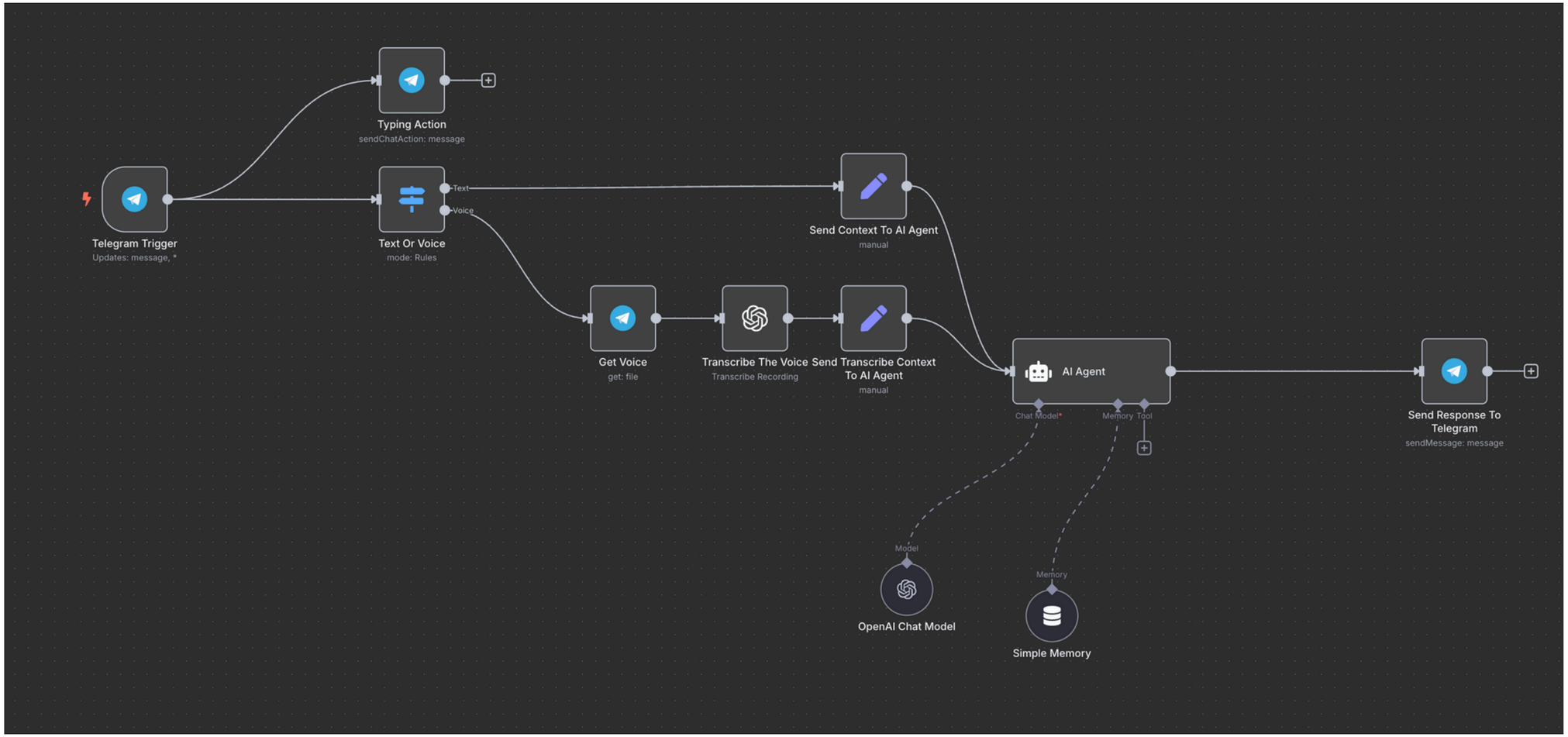
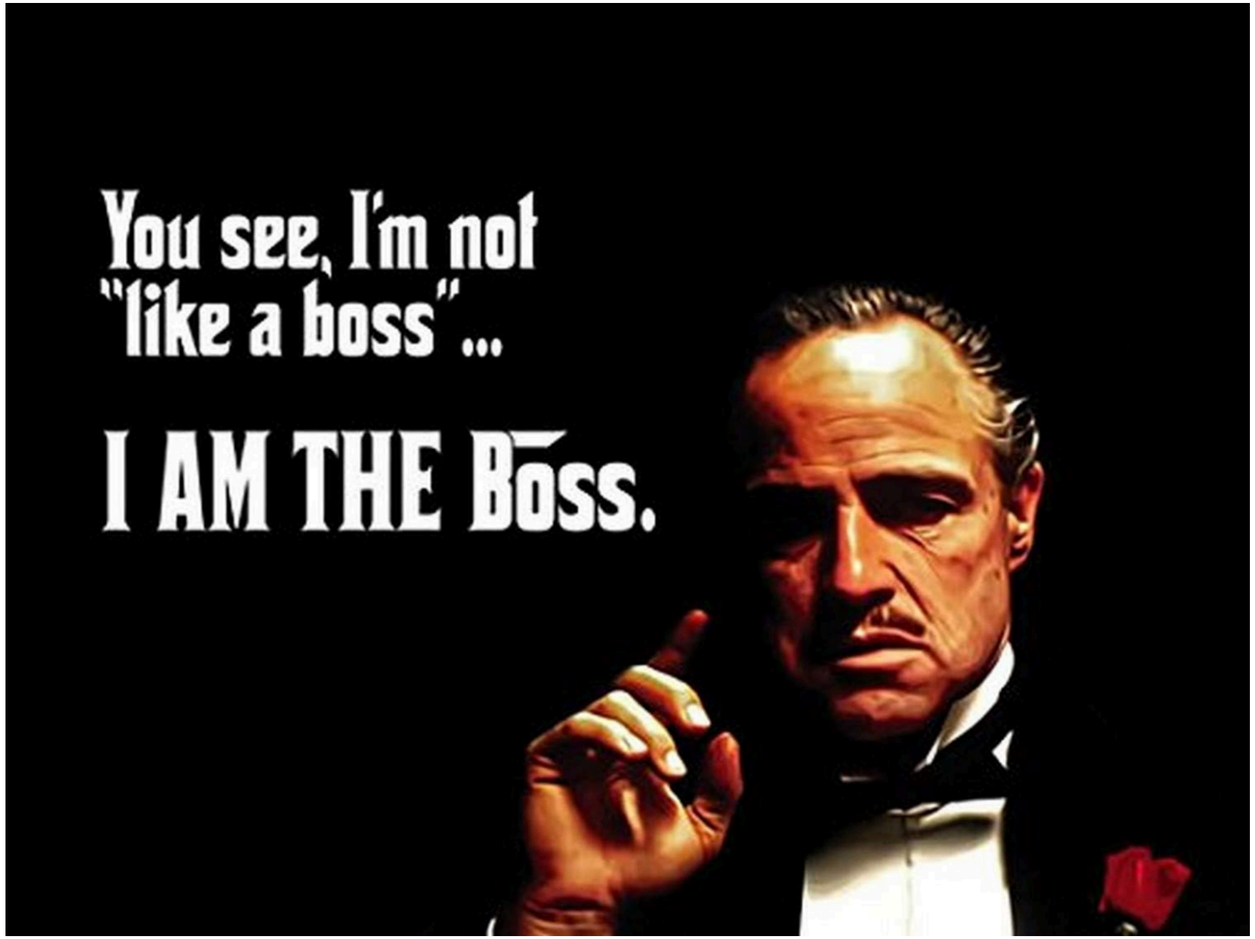
Automation
Rule-Based Predefined Processes Sequential Tasks Limited Decision-Making Requires Human Input to Change

AIAgent
Data-Driven Adaptive Learning Complex Reasoning Autonomous Actions Dynamically Modifiable

Agentic AI Flow 1 - Doubt Solving Assistant



Agentic AI Flow - Telegram Personal Assistant



Steps to create a Telegram Agent